

Homework — 1.3.2 Databases — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] 1 mark: `MemberID` in the `Booking` table is the foreign key. 1 mark: it links to `MemberID`, the primary key of the `Member` table.

2. [3] Sample answer:

```
SELECT FirstName, Surname
FROM Member
WHERE MembershipType = 'Junior'
ORDER BY Surname DESC;
```

1 mark: `SELECT FirstName, Surname FROM Member`; 1 mark: `WHERE MembershipType = 'Junior'`; 1 mark: `ORDER BY Surname DESC` (must include `DESC`).

3. [3] Sample answer:

```
SELECT Member.Surname, Booking.CourtNumber, Booking.BookingDate
FROM Member
JOIN Booking ON Member.MemberID = Booking.MemberID;
```

1 mark: `SELECT` lists `Surname`, `CourtNumber` and `BookingDate`; 1 mark: `FROM`/`JOIN` names both tables; 1 mark: correct `ON Member.MemberID = Booking.MemberID`. (Accept table aliases or an implicit join in the `WHERE` clause.)

4. (a) [2] 1 mark: a single order record would contain a repeating group / multiple products (multiple `ProductID`, `ProductName`, `Quantity` values) in one row, so the values are not atomic. 1 mark: 1NF requires no repeating groups, all attribute values to be atomic (single-valued), and a defined primary key. (Accept: split the products into separate rows/an `OrderLine` table.)

(b) [2] 1 mark: this is a **partial dependency** (a non-key attribute depending on only part of a composite key). 1 mark: it is removed by reaching **Second Normal Form (2NF)**.

5. [2] 1 mark: once a transaction has been committed, its changes are saved permanently. 1 mark: the changes survive a subsequent failure such as a power cut or system crash (e.g. written to non-volatile storage / via a transaction log).

Part B (6 marks)

6. [2] 1 mark: lossy compression (for a typical web photograph, e.g. JPEG). 1 mark for a valid reason, e.g. it gives a much smaller file so it uploads/downloads faster and uses less bandwidth, and slight quality loss is acceptable for viewing. (Accept lossless with valid justification only if the requirement is exact reproduction, but the expected answer for a web photo is lossy.)

7. [2] 1 mark: the Waterfall model is a sequential/linear life cycle where each stage (analysis, design, implementation, testing, maintenance) is completed before the next begins. 1 mark for a drawback, e.g. inflexible / hard to go back and change earlier stages; the customer only sees the product near the end; poor for projects with changing requirements.

8. [2] 1 mark + 1 mark for a developed reason, e.g. an SSD has no moving parts so it is faster (lower access/read times) and less likely to be damaged by being dropped/knocks **[1]**, and it uses less power so improves battery life / is silent / lighter **[1]**. (Any two valid points, or one point well developed.)

Total: 14 + 6 = 20 marks